

```
9. print(comparison)
10.
11. # Calculate the adjustment ratio
12. data['Adjustment Ratio'] = data['Adj Close'] / data['Close']
13.
```

Tesla had a 5-for-1 stock split on August 31, 2020, which would appear as a sudden 80% price drop in unadjusted data. Using adjusted prices ensures your analysis doesn't interpret this corporate action as a market movement. When developing trading algorithms, always use adjusted prices for historical analysis but be aware that your live trading will occur at current market prices.

Resampling Data: Changing Time Frequencies

Financial data comes in various time frequencies (daily, hourly, minute-by-minute), and you often need to convert between them:

```
1. import yfinance as yf
2. import pandas as pd
3.
4. # Download daily data
5. daily_data = yf.download("AAPL", start="2023-01-01", end="2023-07-01")
6.
7. # Resample to weekly data (using last day of each week)
8. weekly_data = daily_data.resample('W').last()
9.
10. # Resample to monthly data (using last day of each month)
11. monthly_data = daily_data.resample('M').last()
12.
13. # For OHLC data, we can use specialized resampling
14. monthly_ohlc = daily_data.resample('M').agg({
15.     'Open': 'first', # First day's open price
16.     'High': 'max',   # Highest price in the month
17.     'Low': 'min',    # Lowest price in the month
18.     'Close': 'last', # Last day's close price
19.     'Volume': 'sum'  # Total volume for the month
20. })
21.
```